

The Eddy Experiment: GNSS-R speculometry for directional sea-roughness retrieval from low altitude aircraft

O. Germain,¹ G. Ruffini,¹ F. Soulat,¹ M. Caparrini,¹ B. Chapron,² and P. Silvestrin³

Received 13 July 2004; revised 3 September 2004; accepted 19 October 2004; published 12 November 2004.

[1] We report the retrieval of sea surface directional mean square slope from Global Navigation Satellite System Reflections (GNSS-R) Delay-Doppler-Map (DDM) data collected during an experimental flight at 1 km altitude. We emphasize the utilization of the entire DDM to more precisely infer ocean roughness directional parameters, and argue that the DDM exhibits the impact of both roughness and scatterer velocity. Obtained estimates are analyzed and compared with Jason-1 measurements, ECMWF numerical weather model outputs and optical data. **INDEX TERMS:** 4275 Oceanography: General: Remote sensing and electromagnetic processes (0689); 4520 Oceanography: Physical: Eddies and mesoscale processes; 4594 Oceanography: Physical: Instruments and techniques; 6959 Radio Science: Radio oceanography; 6969 Radio Science: Remote sensing. **Citation:** Germain, O., G. Ruffini, F. Soulat, M. Caparrini, B. Chapron, and P. Silvestrin (2004), The Eddy Experiment: GNSS-R speculometry for directional sea-roughness retrieval from low altitude aircraft, *Geophys. Res. Lett.*, 31, L21307, doi:10.1029/2004GL020991.

1. Introduction

[2] Several GNSS are presently operational or under development, including the pioneering US Global Positioning System (GPS) and the forthcoming European system, Galileo. All-weather, long-term, stable and precise GNSS L-band signals can be used for bistatic remote sensing of the ocean surface and beyond, an emerging concept known as GNSS-R.

[3] Among several applications, two have been emphasized by the community: sea-surface altimetry [see *Ruffini et al.*, 2004, and references therein] and sea-surface “speculometry” (discussed below), focusing on the statistical properties of sea surface gravity wave slopes. Although this paper addresses the latter, we note the intrinsic capability of GNSS-R for providing co-located measurements of both surface roughness and sea level with high spatio-temporal resolution. Recent scatterometer studies [*Chelton et al.*, 2004] show that such a capability would help to quantify the relationship of velocities in the upper ocean (driven by wind stress forcing) with surface height dynamics.

[4] Thanks to its passive multistatic character, GNSS-R holds the potential to provide an unprecedented spatio-

temporal sampling of the ocean surface. The expected high spatial and temporal measurements can serve operational applications. For instance, like scatterometers, GNSS-R can complement ocean wind and wave models. Such rain immune new data could help quantify atmosphere-ocean coupling, including momentum and energy fluxes under extreme conditions for hurricane modeling. Other scientific applications can be cited, such as sea surface breaking/whitecapping and gas exchange global characterization, important ingredients for understanding the ocean’s biogeochemical response to, and its influence on, climate change. Indeed, a very promising approach to quantify CO₂ flux is to better assess the surface fractional area, which is readily measurable from surface slope measurements [*Gregg et al.*, 2000]. In that case, GNSS-R can provide a more direct measurement to extract gravity surface slope statistical properties and to quantify the role of the ocean in taking up increases of CO₂.

[5] L-band sea surface roughness data can also support L-band radiometric missions such as SMOS (Soil Moisture and Ocean Salinity) and AQUARIUS, i.e., to quantify and separate roughness and salinity contributions to radiometric brightness measurements.

[6] Inferring sea roughness from GNSS-R data requires models for the ocean surface, scattering process and instrument, as well as the choice of a GNSS-R data product to be inverted. There is quite an agreement on the two first aspects in the literature. It has been recognized that the scattering of GNSS signals can be approximated by a Geometric Optics model, where the fundamental physical process is the scattering from mirror-like surface cap elements. This is the reason for our use of the term “speculometry”, stemming from the Latin for mirror, *speculum*: the detected GNSS-R return is dominated by the sea surface slope probability distribution function (PDF)—equivalently, the product of the average number and average curvature of specular points [*Barrick*, 1968]—at scales larger than the electromagnetic wavelength λ .

[7] Assuming gaussianity, three parameters define the L-band slope PDF. The directional mean square slope (DMSS _{λ}) is a symmetric tensor which results from the integration of the ocean energy spectrum at wavelengths larger than λ , and characterizes the ellipsoidal shape of the slope PDF.

[8] We note that DMSS _{λ} has rarely been emphasized as the geophysical parameter of interest. Instead, roughness is typically linked to surface winds—which are thought to be more useful for oceanographic and meteorological users. Unfortunately, the relationship between surface wind and roughness is not one-to-one and requires an additional modeling layer, as it is affected by other factors (e.g., swell, fetch and degree of maturity [see *Gourrion et al.*, 2002]).

¹Starlab, Barcelona, Spain.

²Institut Français pour L’exploitation de la Mer, Plouzané France.

³European Space Research and Technology Centre, European Space Agency, Noordwijk, Netherlands.

[9] Moreover, the product traditionally used for inversion in GNSS-R speculometry is the 1D delay waveform of the reflected signal amplitude, from which the wind speed is inferred assuming an isotropic slope PDF [see, e.g., *Garrison et al.*, 1998, 2002; *Komjathy et al.*, 2000; *Cardellach et al.*, 2003]. Attempts have also been made to estimate the wind direction by fixing the PDF isotropy to some theoretical value (~ 0.7) and using at least two satellite reflections with different azimuths [see, e.g., *Zuffada et al.*, 2003; *Garrison*, 2003; *Komjathy et al.*, 2004].

[10] We work here with a product of higher information content: the 2D DDM of the reflected signal amplitude [see, e.g., *Lowe et al.*, 2002]. As proposed by *Ruffini et al.* [2000], the provision of this extra dimension should allow for the robust estimation of the DMSS $_{\lambda}$ parameters through the direct fitting/estimation of the entire DDM. *Elfouhaily et al.* [2002] give a first order approximation for inversion.

[11] The full DDM fitting technique is used for the first time here to analyze GNSS-R data collected during the Eddy Experiment. This campaign and the altimetric data analysis are reported elsewhere [*Ruffini et al.*, 2004; *Soulat*, 2003], and is only briefly described in Section 2. The retrieval methodology relies on a least-squares fit of the speculometric model, as discussed in the third section. In the fourth, results are compared to ancillary data (Jason-1 radar altimeter, ECMWF numerical weather model, optical data). Finally, another important outcome of the exhaustive exploitation of the information contained in the DDM product is discussed: we evidence that a small part of the Doppler spread can be attributed to the mean “scatterer velocity”, i.e., the rapid motion of the λ (or larger) sized sea-surface scattering *specula*.

2. Data Collection and Pre-processing

[12] The data set (direct/reflected GPS signals with aircraft kinematic data) was gathered in September 2002. The aircraft overflew the Mediterranean Sea, off the coast of Catalonia (Spain), northwards from the city of Barcelona for about 150 km at 1000 m altitude and 45–75 m/s speed. The area is crossed by the ground track #187 of the Jason-1 radar altimeter, which the aircraft overflew during the satellite overpass for comparison. The track was overflown twice: the first time during the ascending pass (from South to North) at low speed (45–60 m/s) and the second during the descending pass at a faster speed (65–75 m/s) due to wind. The time shift between the two passes over a same point on the track ranged from 45 min to 2 h 15 min. During the ascending pass, PRNs 08, 10 and 24 were visible with elevations ranging 30°–85°, while PRNs 08, 10 and 29 were visible during the descending track with elevations 40°–75°. The configuration of this test flight was not optimal for speculometry. From such low altitude the glistening zone is coarsely Delay-Doppler mapped: at nadir, the approximate scales of first-chip, first-Doppler and one-sigma glistening zones are 770, 170 and 330 m respectively.

[13] The raw GPS signals were acquired with a modified TurboRogue receiver, sampled at 20.456 MHz and pre-processed with a dedicated software (STARLIGHT). Correlations were computed at 81 delay lags while the

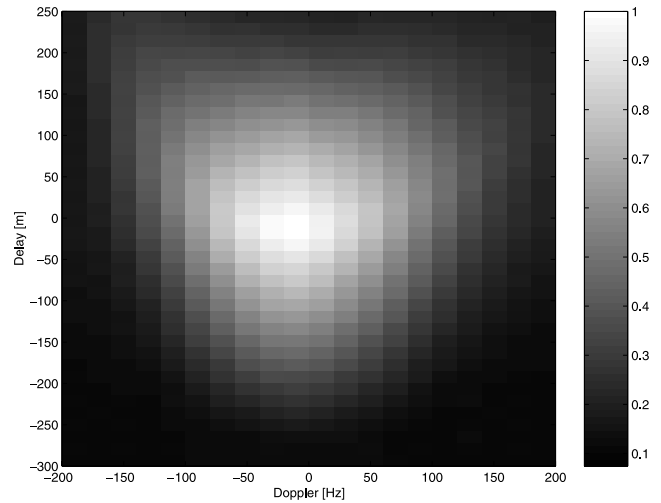


Figure 1. Example of GPS-R Delay-Doppler Map.

Doppler dimension spanned -200 to 200 Hz with a step of 20 Hz. The coherent/incoherent integration times were respectively set to 20 ms and 10 s—the averaged DDM were produced at the rate of 0.1 Hz after summation of 500 incoherent looks (see Figure 1 for a sample).

3. Speculometric Model and DDM Inversion

[14] The link between the DDM mean power at delay-Doppler τ, f and the effective L-band sea-surface slope PDF, $\mathcal{P}(s_x, s_y)$, is given by [see *Zavorotny and Voronovich*, 2000]

$$P(\tau, f) = \int dx dy \frac{G_r}{R_t^2 R_r^2} \cdot \frac{q^4}{q_z^4} \cdot \mathcal{P}\left(\frac{-q_x}{q_z}, \frac{-q_y}{q_z}\right) \cdot \chi^2[\tau_m(x, y) - \tau_c - \tau, f_m(x, y) - f_c - f], \quad (1)$$

where G_r is the receiver antenna pattern, R_t and R_r the distances from generic point on sea-surface to transmitter and receiver, (q_x, q_y, q_z) the scattering vector, χ the Woodward Ambiguity Function (WAF), $\tau_m(x, y)$ and $f_m(x, y)$ the delay-Doppler coordinates on the sea-surface and (τ_c, f_c) the delay/Doppler offset of the geometric specular-point with respect to the direct signal (the DDM “center”). Accounting for the receiver mean thermal noise P_N and including a scaling parameter α , the mean amplitude of the DDM can be written as $A(\tau, f) = \sqrt{\alpha P(\tau, f) + P_N}$. As discussed above, $\mathcal{P}$ is described by the DMSS $_{\lambda}$ parameter set, which defines an elliptic quadratic form in slope space. The ellipse major and minor principal axes are often called MSS up-wind (mss_u) and MSS cross-wind (mss_c) respectively. In the following, we will refer to the Total MSS ($MSS_{tot} = 2\sqrt{mss_u \cdot mss_c}$, proportional to the ellipse area and directly related to nadir σ°), the Slope PDF azimuth (SPA, the direction of semi-major axis with respect to North) and the Slope PDF Isotropy (SPI, equal to mss_c/mss_u).

[15] The inversion was performed through minimization of the mean square difference between model and data DDMs. The main difficulty stemmed from the presence of several nuisance parameters in the forward model (mainly τ_c and f_c but also α). The DDM centers were affected by the aircraft trajectory (altitude and vertical velocity) to first

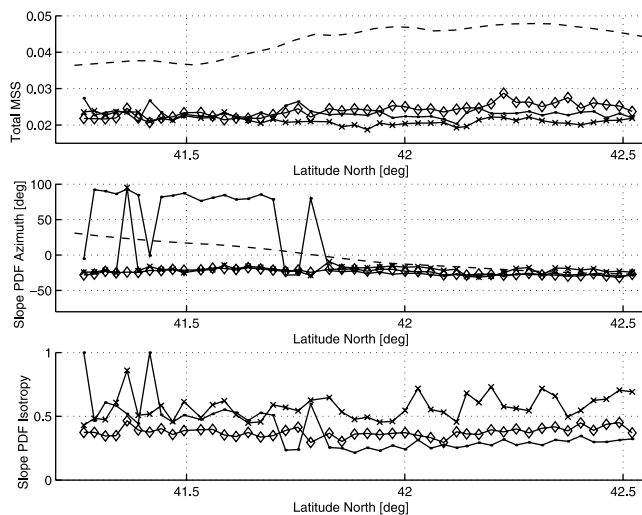


Figure 2. $DMSS_{\lambda}$ estimation with the DDM least-square inversion approach, along the descending (North to South) track—MSS top, SPA middle and SPI bottom (dot: PRN08, diamond: PRN10, cross: PRN29). The total Ku-Band MSS measured by Jason-1 and the ECMWF wind direction are also shown (dashed line).

order but also by geophysical parameters (such as sea level). They needed to be accurately known in order to estimate $DMSS_{\lambda}$. For this reason, the $DMSS_{\lambda}$ and nuisance parameters were jointly estimated in an iterative manner.

4. Results and Analysis

[16] The values of $DMSS_{\lambda}$ estimated along the descending track of the flight are shown on Figure 2. The top plot illustrates the variations of MSS_{tot} . Inter-PRN consistency is reasonable in the south part of the track but worsens slightly in the north part. For comparison, the total MSS in Ku-band was derived from the Jason-1 σ^0 co-located measurements at 1 Hz sampling (7 km) and 20 km resolution. The Jason-1 MSS was obtained through the simple relationship $MSS = \kappa/\sigma^0$, κ being the effective (empirical) Fresnel coefficient, here set to 0.45. As expected, the level and dynamic of MSS decreased with longer wavelength (from 2 cm in Ku-band to 19 cm in L-band). The lower dynamic of L-band MSS impeded a clear trend comparison, although the measurements of PRN-10 seem in good agreement with Jason-1. Jason-1 wind speed, derived from both the Ku-band σ^0 and the significant wave height (~ 2 m), ranged from 9 to 13 m/s along the track. Translating this wind speed into L-band MSS through the spectrum of *Elfouhaily et al.* [1997] yields values between 0.0220 and 0.0255, in-line with GNSS-R results. We note, however, that the assumption of a wind-driven spectrum was not really warranted during the campaign.

[17] SPA estimation results are presented on the middle plot. There is good inter-PRN consistency and the apparent discrepancy in the south part of the track can be attributed to estimation degeneracy. DDM inversion for SPA is degenerate in at least two cases: zenith transmission and when the receiver moves towards the transmitter. Then, the Delay-Doppler lines are symmetric about the receiver direction and

it becomes impossible to distinguish a slope PDF from its mirror image about the receiver direction. This effect is observed here with two SPA minima (-20° and 80°) symmetric about the aircraft heading direction (30°). In this part of the track, the azimuth of PRN-08 is about 50° , almost aligned with the aircraft heading direction. In the north part of the track, SPA estimates match very well ECMWF wind direction. In the south part, the mismatch reaches up to 50° . However, we underline that surface wind is not the only element driving SPA and that swell is likely to have contributed.

[18] Finally, the bottom plot shows the SPI variations along the track. The reflected signals are strongly directional. The wind-driven spectrum of *Elfouhaily et al.* [1997] for a mature sea predicts a SPI value around 0.65, largely insensitive to wind speed. The apparent significant departure from this reference value is a probable signature of an under-developed sea with the presence of swell. However, the relatively poor consistency among PRNs remains an issue: further work is needed to validate the accuracy of these SPI estimates and to better understand the potential information content of this product.

[19] As a second new outcome of this work, we now discuss the signature of “scatterer velocity” in the data, i.e., the signature of fast moving *specula* with size (curvature) larger than ~ 20 cm. Such a signature can be detected when comparing the total MSS along the ascending and descending tracks, as estimated by the least-squares approach: a drastic discrepancy (up to 33%) was observed for two passes shifted by less than one hour over the same track point.

[20] Multipath effects could conceivably lead to a Doppler width modulation (corrupting MSS estimation). However, some azimuth dependence should have been observed and was not. Another possible cause could be a changing aircraft attitude, but the aircraft roll and pitch values were checked to be always nominal. We concluded that the most likely explanation was a geophysical Doppler width modulation by the overall sea surface dynamics as detected from the mean *speculum* motion. While our inversion approach assumed a “frozen” surface, the relative velocity between receiver and scatterer should be taken into account for proper inversion. At high receiver velocities the assumption is sound—the scatterer velocity impact will then not be significant—but marginal at low speeds. This analysis is consistent with the fact that the MSS estimated in the ascending track (slower ground speed, 45–60 m/s) showed abnormal high values compared to the ones estimated during the ascending track (faster, 65–75 m/s).

[21] In order to test this simply, we exploited a first order relationship between the moments of the DDM and the full set of $DMSS_{\lambda}$ (ignoring the impact of the bistatic WAF and antenna): $MSS = \lambda^2 B^2 / (2V^2 \sin^2 \epsilon)$, where ϵ is the transmitter elevation, V the receiver speed and B the DDM Doppler bandwidth [see *Elfouhaily et al.*, 2002]. Assuming no significant MSS variation between the times of the ascending and descending passes and considering the relative velocity between *specula* and aircraft, i.e., $V \pm v_s$ for the descending/ascending passes respectively, we solved for the mean *speculum* velocity v_s (for simplicity assumed parallel to the ground track). Good correlation between the inferred *specula* speed and Jason-1 wind speed can be observed in Figure 3. Moreover, from the dispersion relation, an average

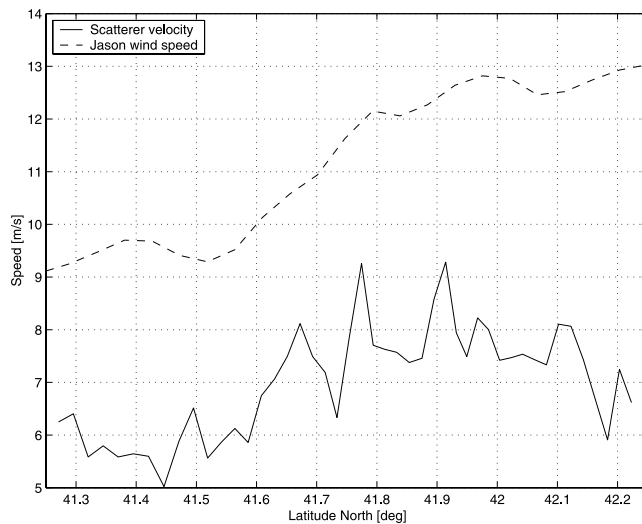


Figure 3. Average scatterer velocity obtained from the first order MSS model and Jason-1 wind speed.

scatterer speed of 8 m/s corresponds to waves of about 45 m wavelength. The detected motion thus seems to be associated with the longer and more coherent wave components, consistent with optical observations of the swell vector (wavelength and direction [see Soulat, 2003]), revealing the presence of a northerly (almost aligned with the flight track) generated 48 m wave system. As hypothesized, *specula* contributing the most to the bistatic reflected signals would, on average, coherently travel with the dominating longer wave peak component.

5. Conclusion

[22] We reported the inversion of GNSS-R signals using, for the first time, the full DDM for the retrieval of the sea-surface directional mean square slope, $DMSS_{\lambda}$; the estimates show good inter-PRN consistency (except for the measured anisotropy SPI) and reasonable agreement with other sources of data.

[23] Analysis of the DDM has revealed a geophysical signature associated with the mean sea surface scatterer velocity: *specula* appear to travel on longer waves with velocities that can reach 5–10 m/s, impacting significantly the Doppler bandwidth observed on slow-moving receivers (e.g., airborne or ground-based, Soulat *et al.* [2004]). This geophysical signature opens new opportunities for GNSS-R speculometry: to infer either $DMSS_{\lambda}$ or a combination of $DMSS_{\lambda}$ and scatterer velocity, depending on the aircraft speed. Further research will take into account the correct deformation of the Doppler lines on the surface for the search of a scatterer mean velocity vector. Finally, we emphasize that the flight was not optimized for speculometry: higher and faster flights are needed to consolidate the DDM inversion technique and to test new higher resolution inversion concepts.

[24] **Acknowledgments.** The experimental campaign and data analysis were respectively carried out under the ESA contracts TRP-ETP-137A (PARIS-Gamma) and 3-10120/01/NL/SF (OPSCAT). We thank the Institut Cartogràfic de Catalunya for flawless flight operations. All Starlab authors

have contributed significantly; the Starlab author list has been ordered randomly.

References

- Barrick, D. E. (1968), Rough surface scattering based on the specular point theory, *IEEE Trans. Antennas Propag.*, 16(4), 449–454.
- Cardellach, E., G. Ruffini, D. Pino, A. Rius, A. Komjathy, and J. Garrison (2003), Mediterranean balloon experiment: GPS reflection for wind speed retrieval from the stratosphere, *Remote Sens. Environ.*, 88, 351–362.
- Chelton, D. B., M. G. Schlax, M. H. Freilich, and R. F. Milliff (2004), Satellite measurements reveal persistent short-scale features in ocean winds, *Science*, 303(5660), 978–983.
- Elfouhaily, T., B. Chapron, K. Katsaros, and D. Vandemark (1997), A unified directional spectrum for long and short wind-driven waves, *J. Geophys. Res.*, 102(C7), 15,781–15,796.
- Elfouhaily, T., D. Thompson, and L. Linstrom (2002), Delay-Doppler analysis of bistatically reflected signals from the ocean surface: Theory and application, *IEEE Trans. Geosci. Remote Sens.*, 40(3), 560–573.
- Garrison, J. L. (2003), Anisotropy in reflected GPS measurements of ocean winds, paper presented at 2003 Workshop on Oceanography With GNSS-R, Starlab, Barcelona, Spain, July. Available at <http://starlab.es/gnssr2003/proceedings/>.
- Garrison, J. L., S. J. Katzberg, and M. I. Hill (1998), Effects of sea roughness on bistatically scattered range coded signals from the Global Positioning System, *Geophys. Res. Lett.*, 25(13), 2257–2260.
- Garrison, J. L., A. Komjathy, V. U. Zavorotny, and S. J. Katzberg (2002), Wind speed measurement using forward scattered GPS signals, *IEEE Trans. Geosci. Remote Sens.*, 40(1), 50–65.
- Gourrion, J., D. Vandemark, S. Bailey, and B. Chapron (2002), Investigation of C-band altimeter cross section dependence on wind speed and sea state, *Can. J. Remote Sens.*, 28(3), 484–489.
- Gregg, W. W., M. J. Behrenfeld, F. E. Hoge, W. E. Esaias, N. E. Huang, S. R. Long, and C. R. McClain (2000), NASA/GSFC research activities for the global ocean carbon cycle: A prospectus for the 21st century, *NASA Tech. Memo. 2000-209882*, 22 pp., NASA Goddard Space Flight Cent., Greenbelt, Md.
- Komjathy, A., V. Zavorotny, P. Axelrad, G. H. Born, and J. L. Garrison (2000), GPS signal scattering from sea surface: Wind speed retrieval using experimental data and theoretical model, *Remote Sens. Environ.*, 73, 162–174.
- Komjathy, A., M. Armatys, D. Masters, P. Axelrad, V. U. Zavorotny, and S. Katzberg (2004), Retrieval of ocean surface wind speed and wind direction using reflected GPS signals, *J. Atmos. Oceanic Technol.*, 21(3), 515–526.
- Lowe, S. T., P. Kroger, G. Franklin, J. L. LaBrecque, J. Lerma, M. Lough, M. R. Marcin, R. J. Muellerschoen, D. Spitzmesser, and L. E. Young (2002), A delay/Doppler-mapping receiver system for GPS reflection remote sensing, *IEEE Trans. Geosci. Remote Sens.*, 40(5), 1150–1163.
- Ruffini, G., J. L. Garrison, E. Cardellach, A. Rius, M. Armatys, and D. Masters (2000), Inversion of GPS-R delay-Doppler mapping waveforms for wind retrieval, paper presented at IGARSS 2000, Inst. of Electr. and Electron. Eng., Honolulu. Slides available at <http://starlab.es/staff/giuliospapers/igarss2000.ps.gz>.
- Ruffini, G., F. Soulat, M. Caparrini, O. Germain, and M. Martin-Neira (2004), The Eddy Experiment: Accurate GNSS-R ocean altimetry from low altitude aircraft, *Geophys. Res. Lett.*, 31(12), L21306, doi:10.1029/2004GL019994.
- Soulat, F. (2003), Sea surface remote-sensing with GNSS and sunlight reflections, Ph.D. thesis, Univ. Politèc. de Catalunya/Starlab, Barcelona, Spain. Available at <http://starlab.es/library.html>.
- Soulat, F., M. Caparrini, O. Germain, P. Lopez-Dekker, M. Taani, and G. Ruffini (2004), Sea state monitoring using coastal GNSS-R, *Geophys. Res. Lett.*, 31, L21303, doi:10.1029/2004GL020680.
- Zavorotny, V., and A. Voronovich (2000), Scattering of GPS signals from the ocean with wind remote sensing application, *IEEE Trans. Geosci. Remote Sens.*, 38(2), 951–964.
- Zuffada, C., T. Elfouhaily, and S. Lowe (2003), Sensitivity analysis of wind vector measurements from ocean reflected GPS signals, *Remote Sens. Environ.*, 88, 341–350.

M. Caparrini, O. Germain, G. Ruffini, and F. Soulat, Research Department, Starlab, C. de l'Observatori, s/n, E-08035 Barcelona, Spain. (giulio.ruffini@starlab.es)

B. Chapron, Institut Français pour L'exploitation de la Mer, Plouzané, France.

P. Silvestrin, European Space Research and Technology Centre, European Space Agency, Noordwijk, Netherlands.